VEHICLE DYNAMICS AND CONTROL- Lateral Model

CAR SEGMENT - Sports Car

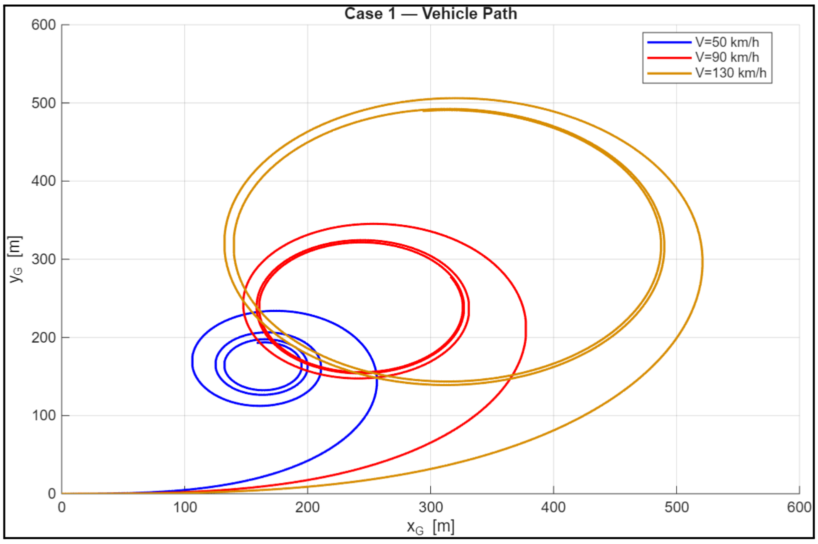
PART-1 OPEN LOOP MANEUVERS

A. **Steady State Steering Pad**

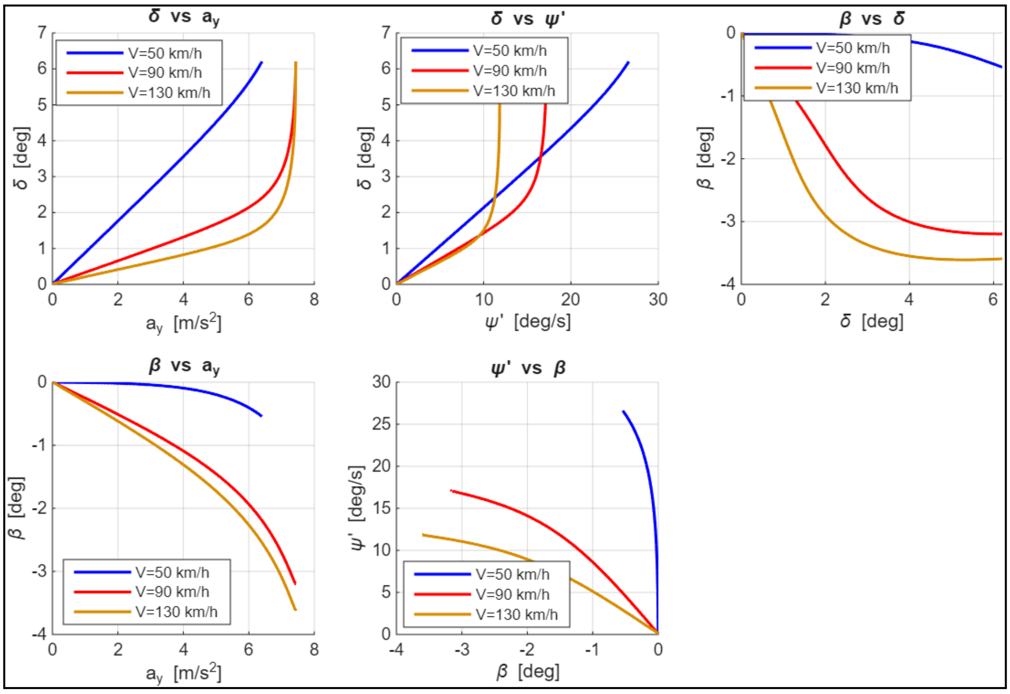
Case 1: Kroll = 0.5

V(km/h) = [50, 90, 130]

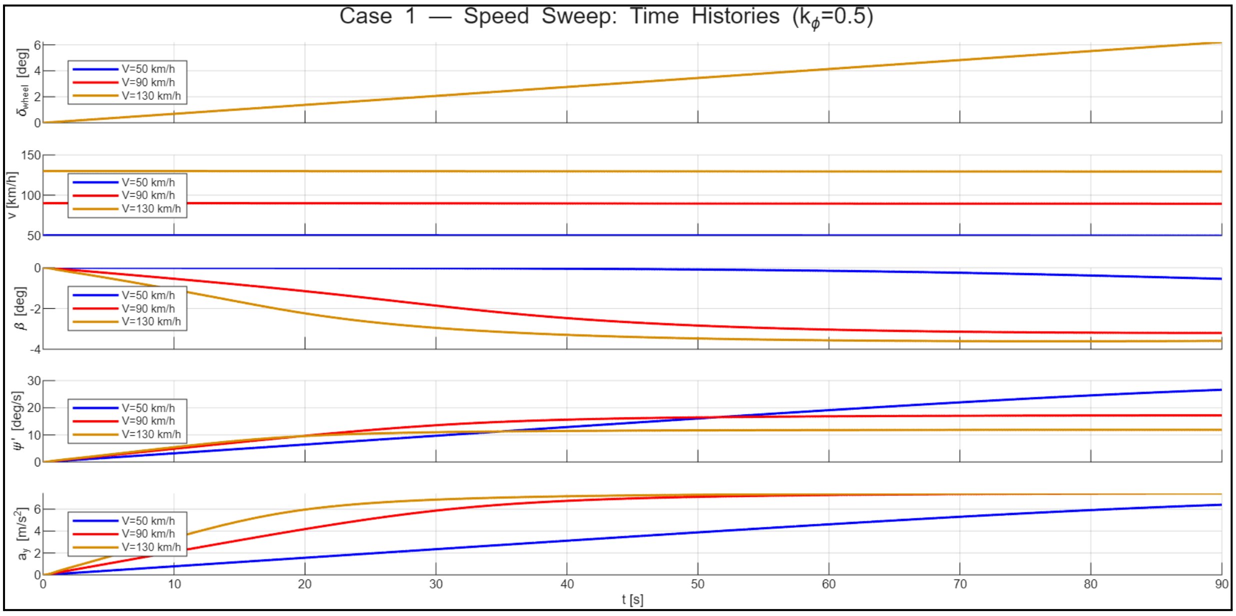
- **Vehicle Path**



- **Lateral Response Characteristics**



- **Time Histories**

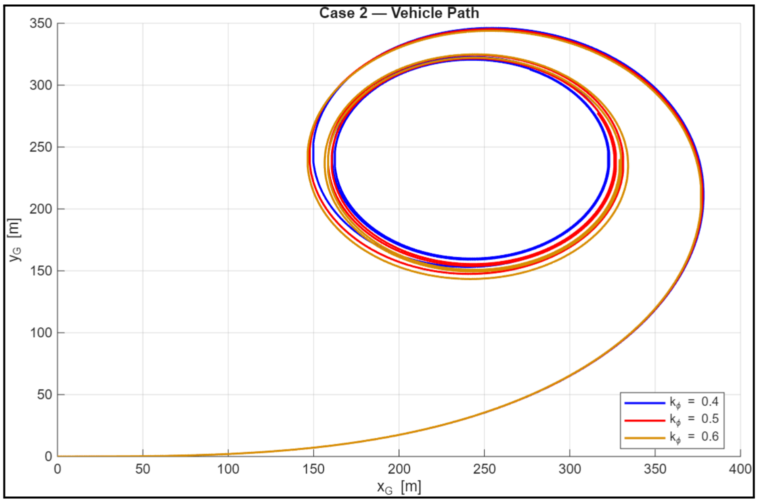


The results show that increasing vehicle speed significantly increases the lateral response of the vehicle for the same steering input. As speed increases from 50 km/h to 130 km/h, the vehicle develops larger lateral acceleration, yaw rate, and sideslip angle, resulting in a progressively tighter and more unstable trajectory.

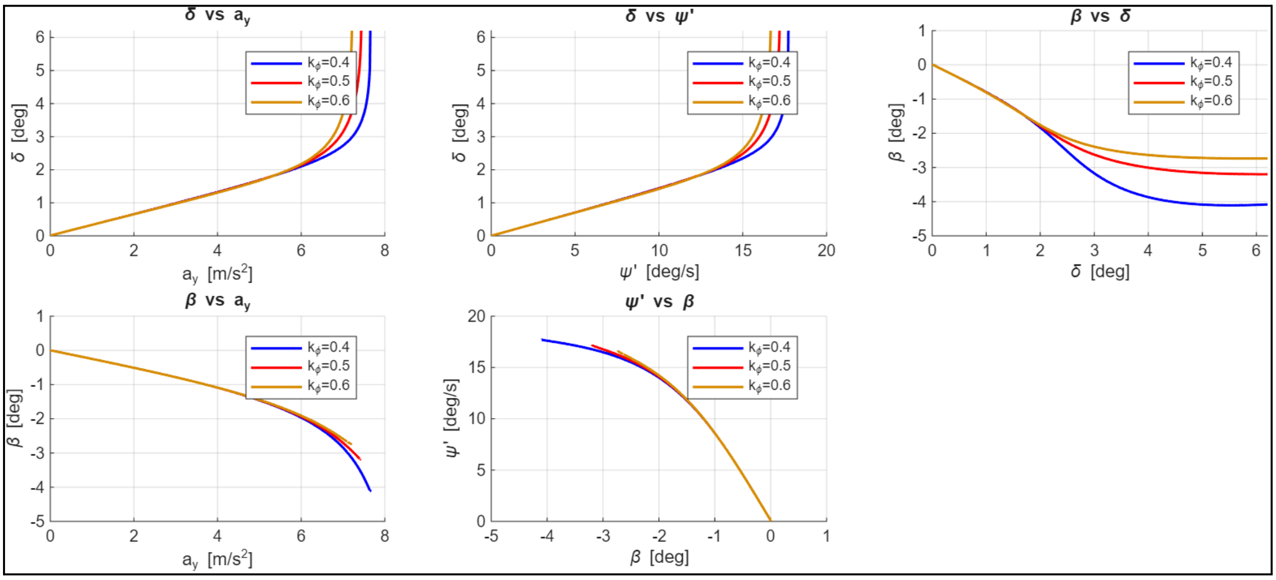
Case 2: V = 90km/h

kroll = [0.4 0.5 0.6]

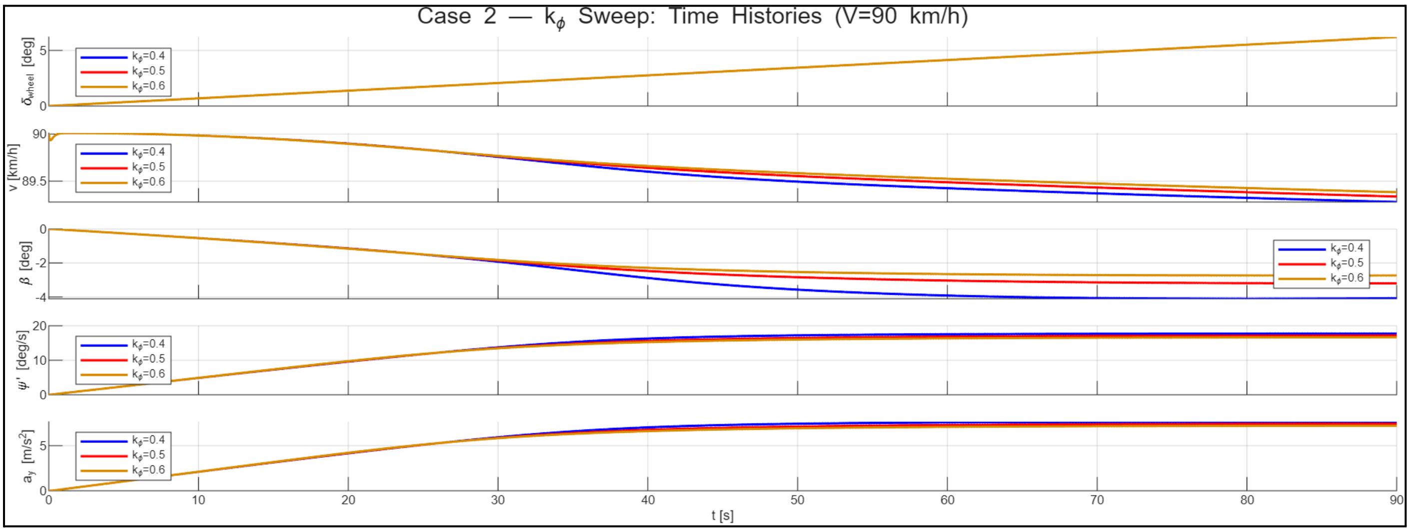
- **Vehicle Path**



- **Lateral Response Characteristics**



- **Time Histories**



The results indicate that increasing the understeer gradient (Kphi) reduces the vehicle's responsiveness to steering inputs. Higher values of Kphi result in smaller sideslip angles, lower yaw rates, and a larger turning radius for the same steering input, indicating a more stable but less agile vehicle. Conversely, lower Kphi values produce a more responsive vehicle with higher lateral response and tighter cornering behaviour.

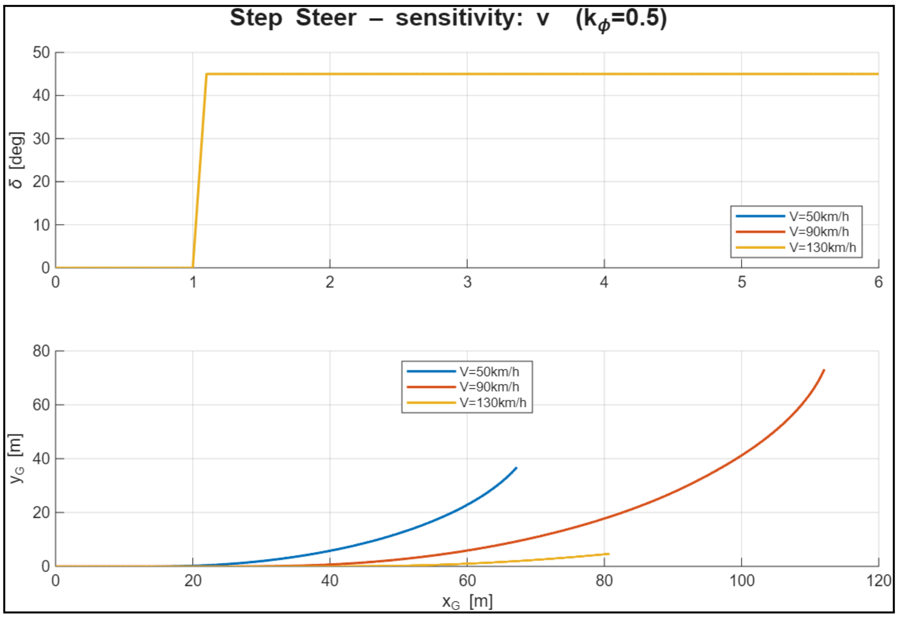
The influence of vehicle speed is significantly stronger than the influence of Kphi. Increasing speed causes substantial increases in lateral acceleration, yaw rate, and sideslip angle, leading to pronounced changes in vehicle trajectory and handling balance. In contrast, varying Kphi primarily modifies the distribution of the cornering response, affecting the degree of understeer and steering sensitivity while producing comparatively smaller changes in the overall vehicle motion. Therefore, speed governs the magnitude of the dynamic response, whereas Kphi mainly determines the handling character and stability of the vehicle.

B. TRANSIENT MANEUVERS

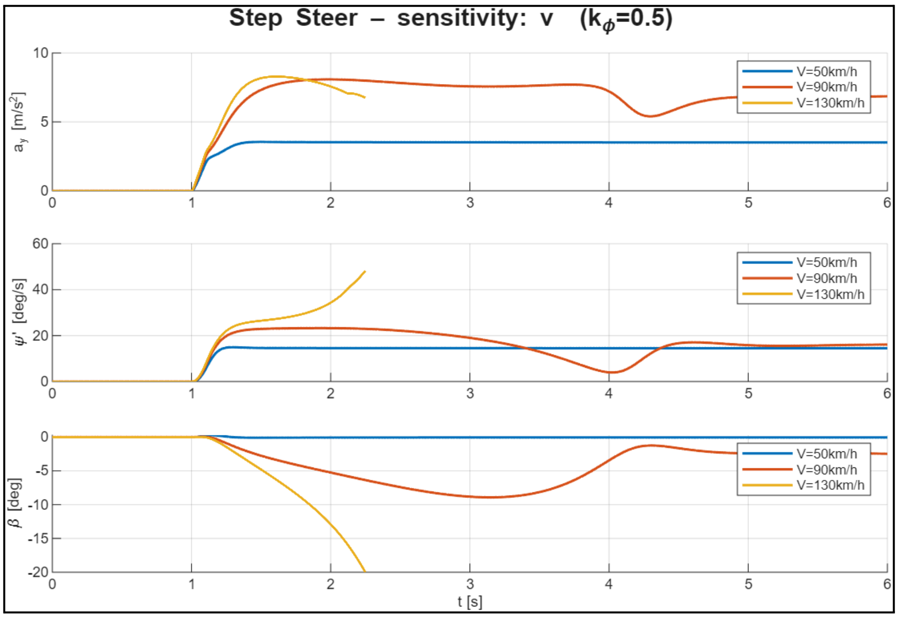
Case 1: kroll = 0.5
V(km/h) = [50 90 130]

i. Step Steer

- Vehicle Path and Input for constant kroll

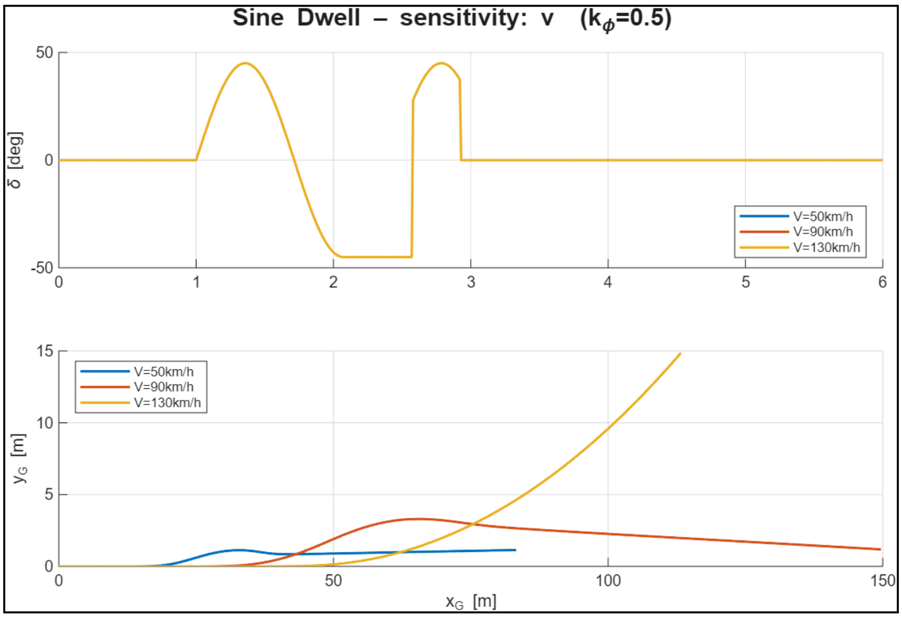


- Time Histories

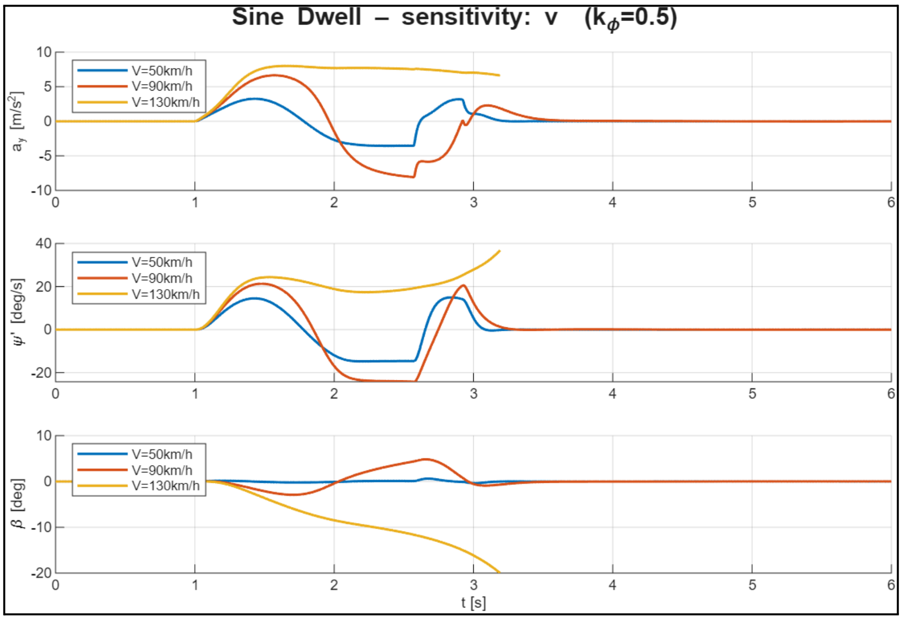


ii. Sine Dwell

- Vehicle Path and Input



- Time Histories



The step steer results show that increasing vehicle speed leads to a stronger dynamic response. Higher speeds generate larger lateral acceleration, yaw rate, and sideslip angle, causing the vehicle to become less stable and more prone to sideslip. The vehicle path also deviates more significantly as speed increases, highlighting the greater sensitivity of the vehicle to steering inputs at high speeds.

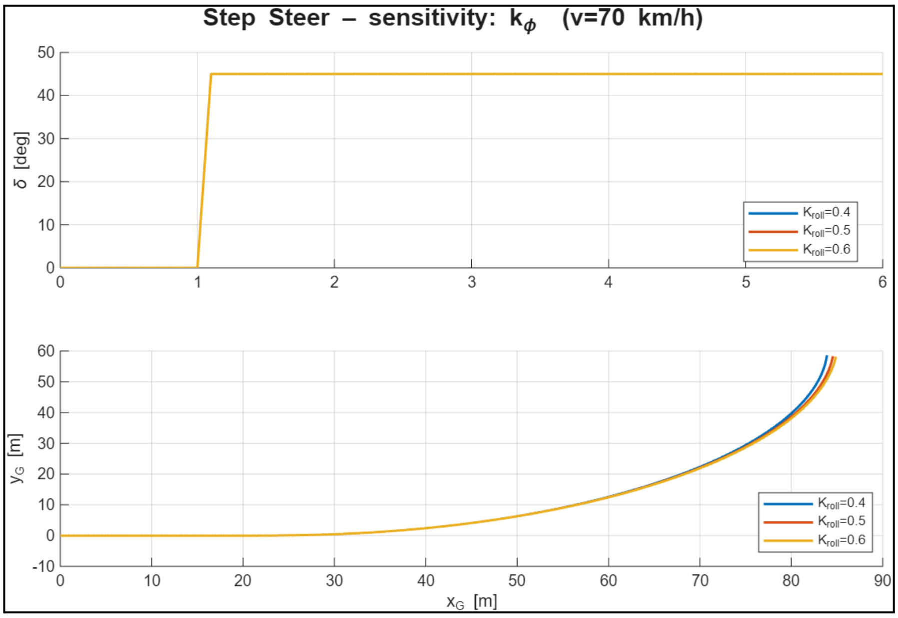
For the sine dwell manoeuvre, increasing speed amplifies the transient vehicle response. Higher velocities result in larger peaks in lateral acceleration, yaw rate, and sideslip angle, with the vehicle exhibiting greater oscillations and a larger lateral displacement. This indicates reduced stability margins and increased dynamic sensitivity during rapid steering reversals at higher speeds.

Case 2: kroll = [0.4 0.5 0.6]

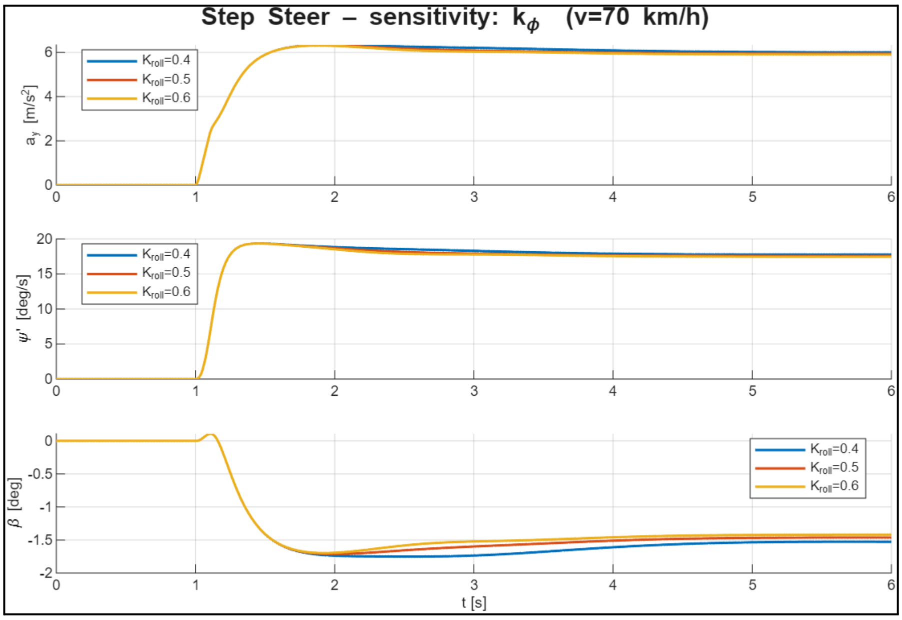
V(km/h) =70 km/h

i. Step Steer

- Vehicle Path and Input

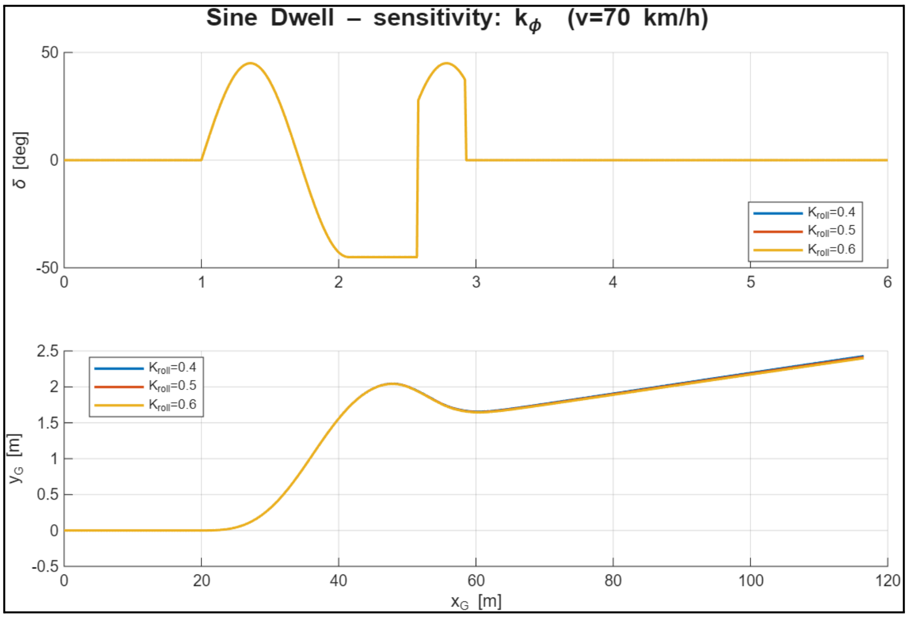


- Time Histories

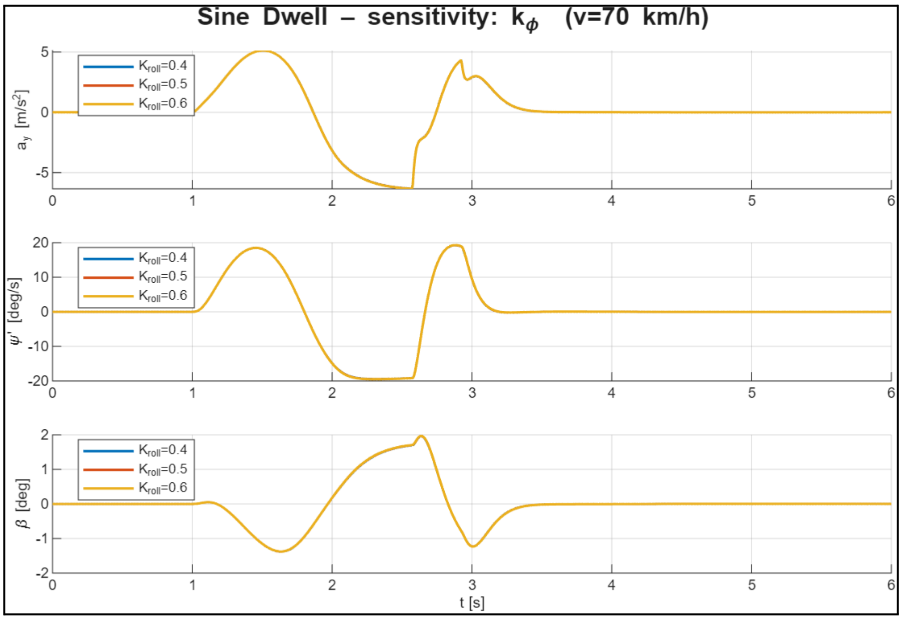


ii. Sine Dwell

- Vehicle Path and Input



- Time Histories



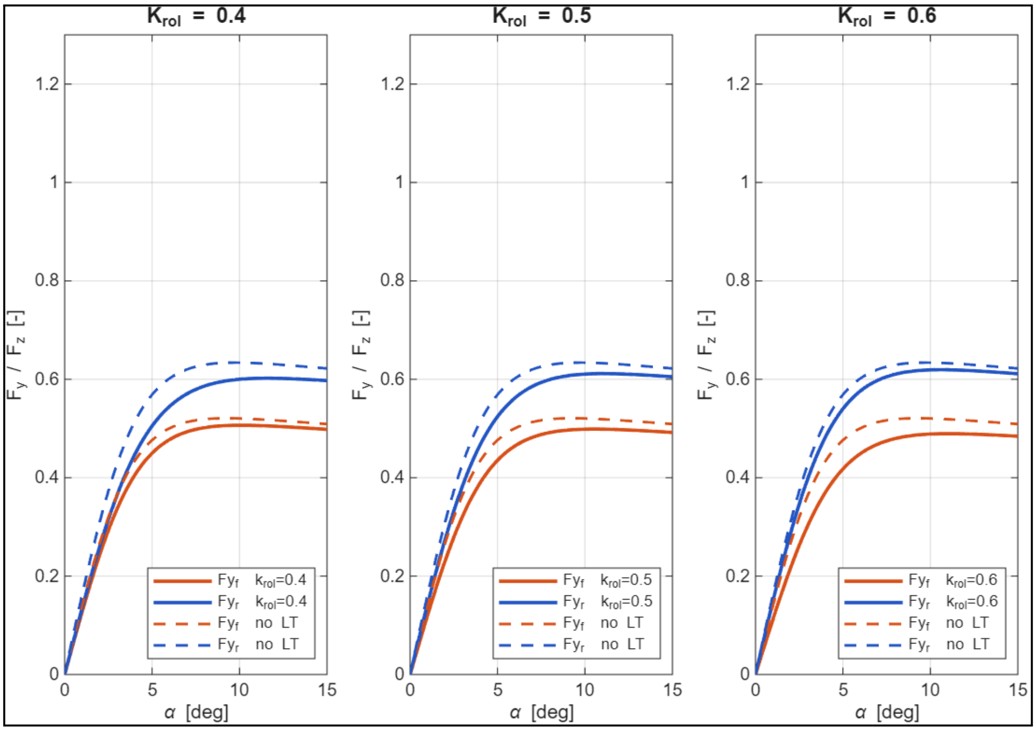
The step steer results show that varying Kphi has a relatively small effect on the overall vehicle response compared to vehicle speed. Increasing Kphi slightly reduces the sideslip angle and yaw response, producing a more stable and less responsive vehicle. The vehicle path and lateral acceleration remain largely unchanged, indicating that the steady-state cornering behaviour is only moderately affected by Kphi variations.

For the sine dwell manoeuvre, changes in Kphi produce only minor differences in the transient response. Higher values of Kphi slightly reduce the peak yaw rate and sideslip angle, resulting in a more damped and stable response during steering reversals. The vehicle trajectory and lateral acceleration remain very similar across all cases, showing that Kphi primarily influences handling balance rather than the overall magnitude of the response.

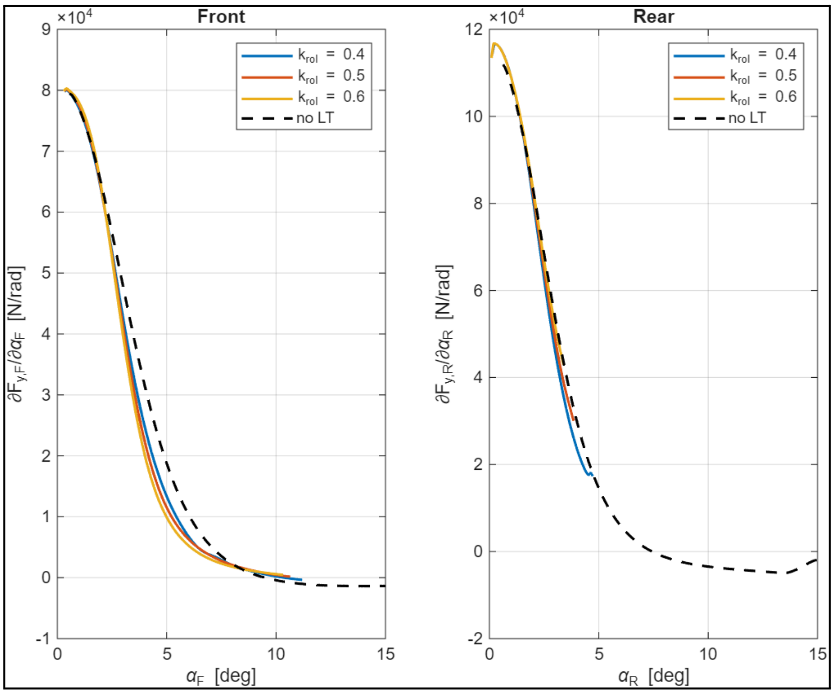
PART 2- AXLE CHARACTERISTICS AND STABILITY

1. Effective Axle Characteristic

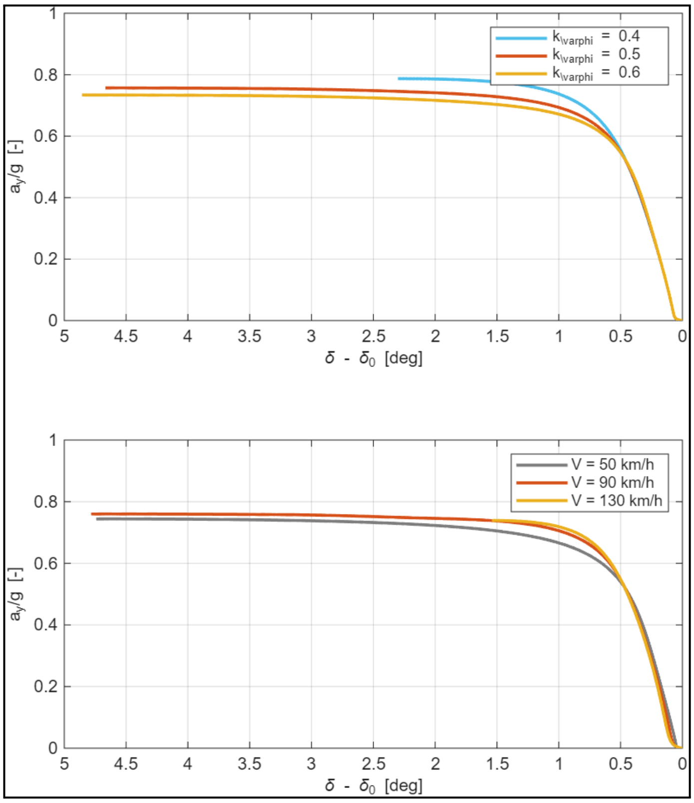
- Fy vs alpha



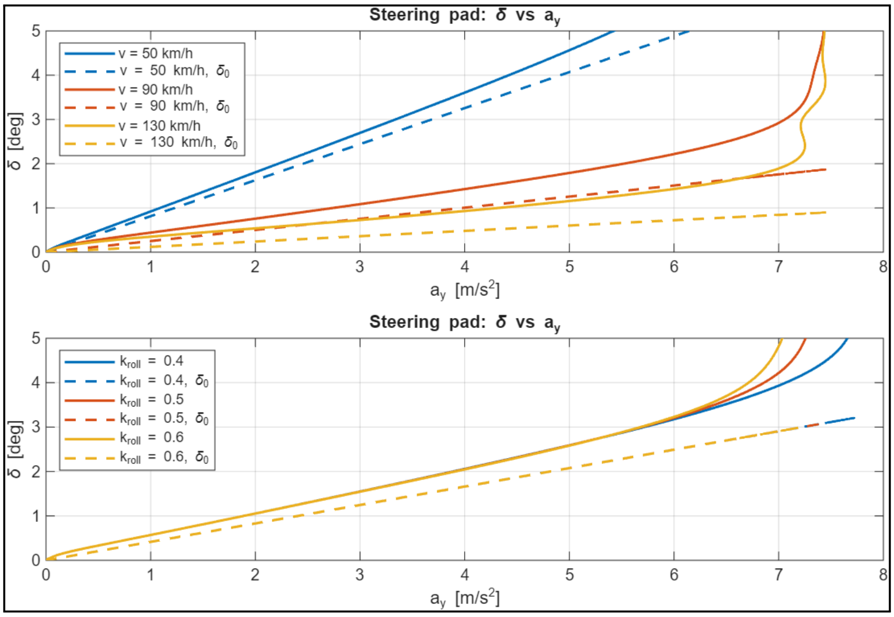
- Derivative of Fy vs alpha



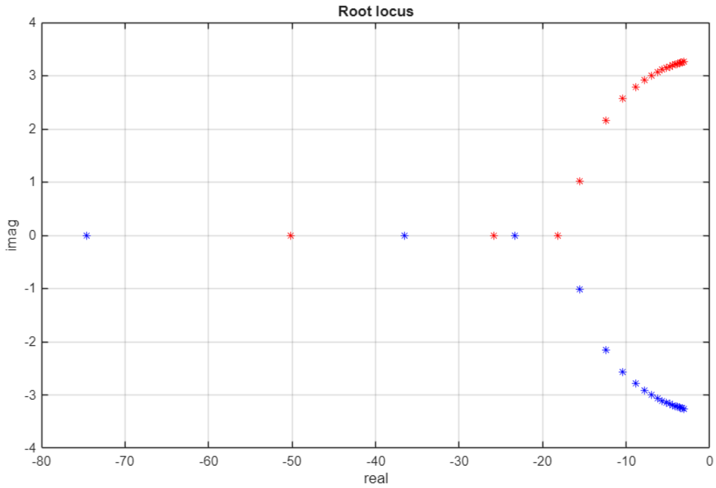
- Handling Diagrams

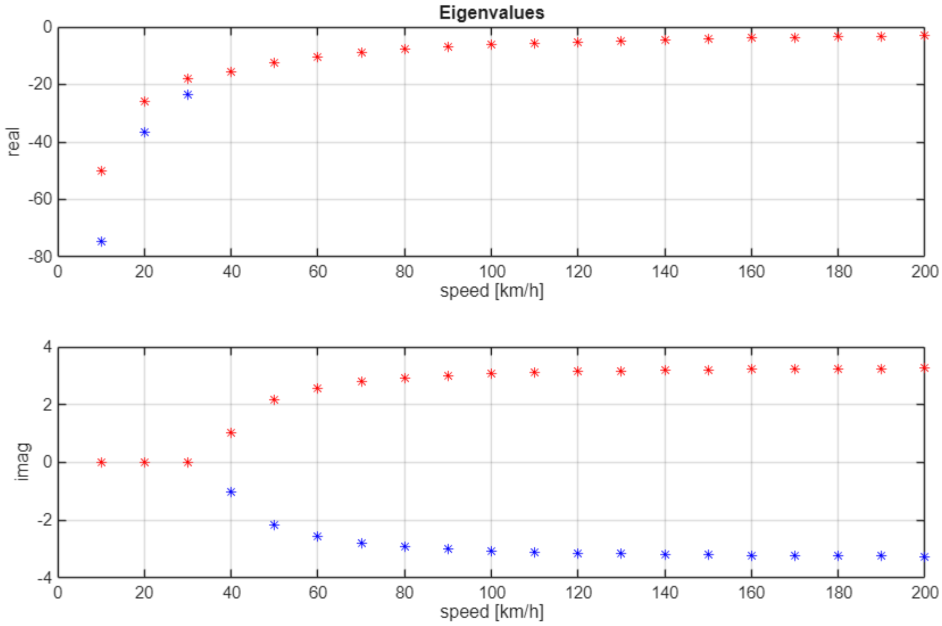


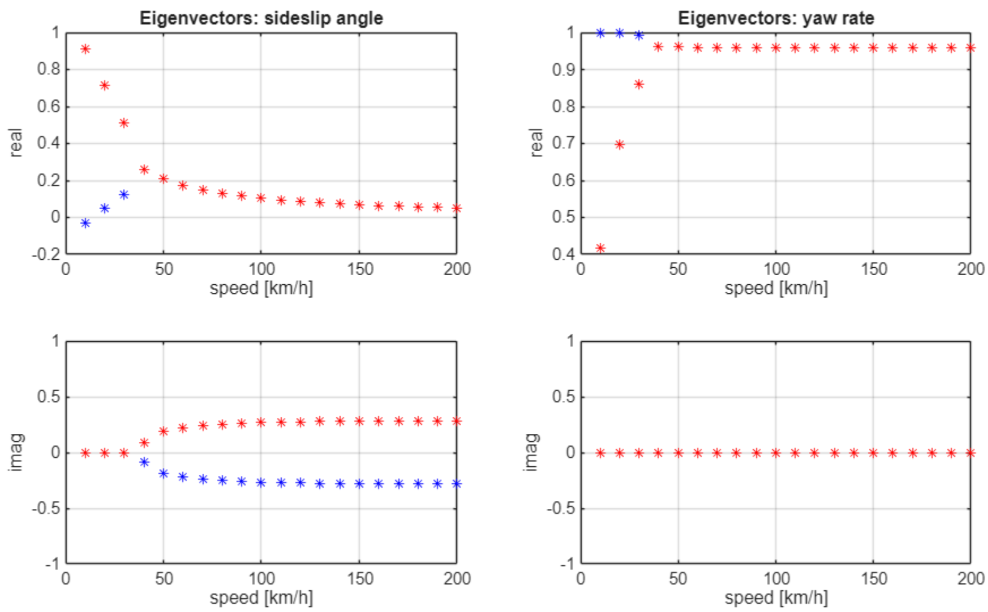
2. δ vs ay of steering pad simulations comparing the results with Ackermann steering geometry



3. Stability Analysis

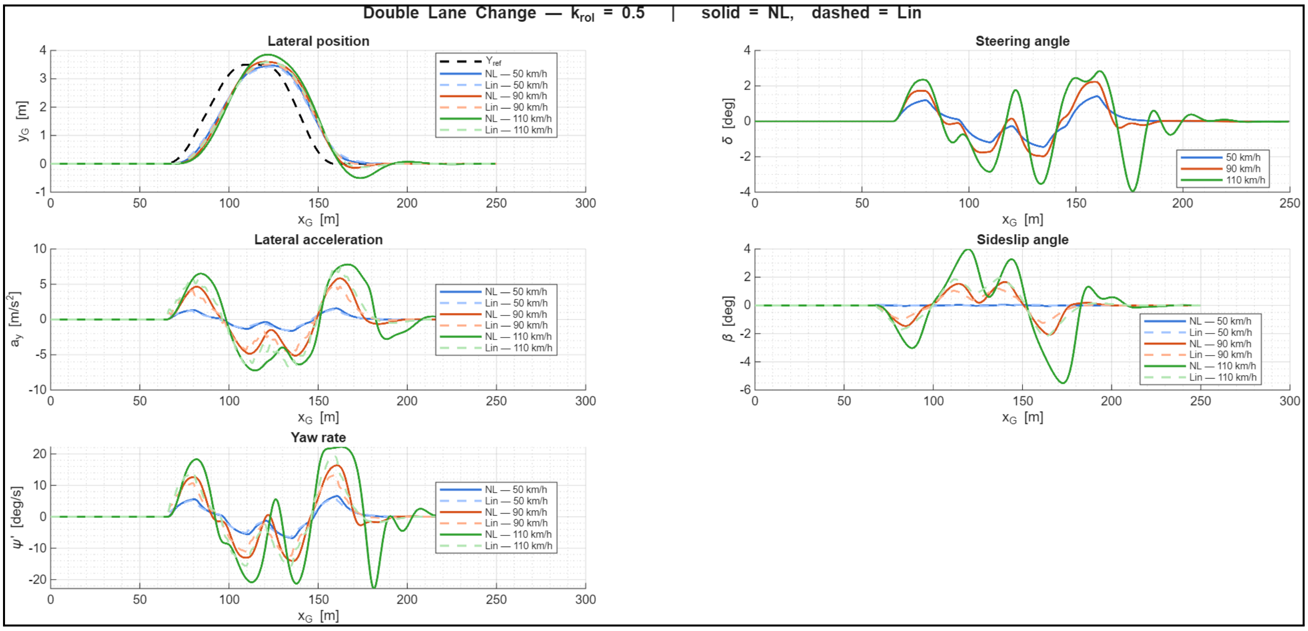






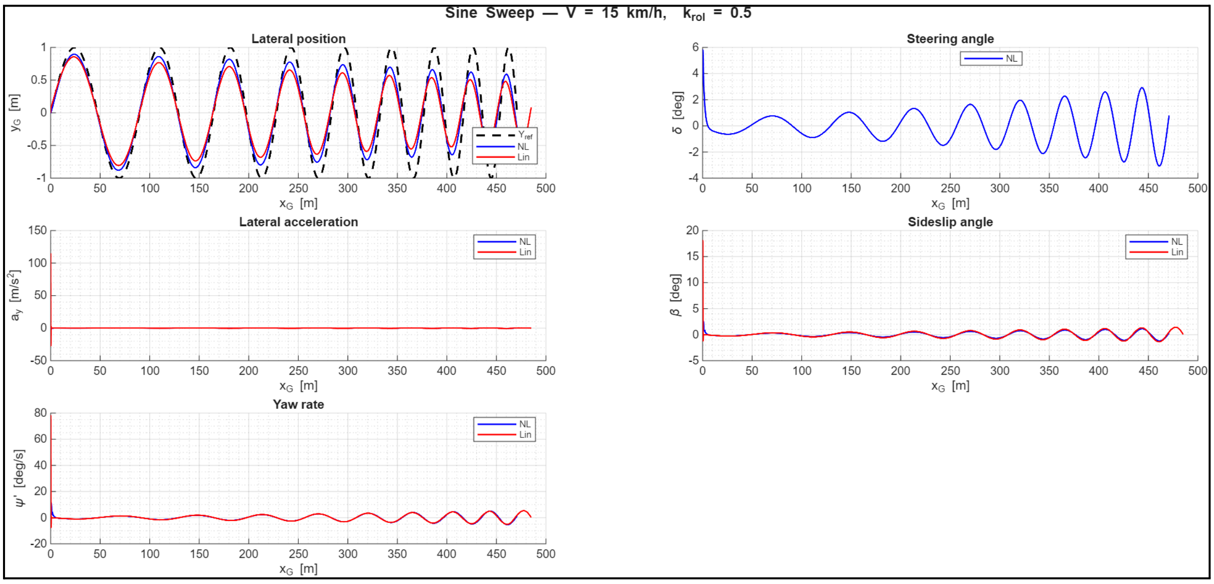
PART 3- PATH FOLLOWER MODEL

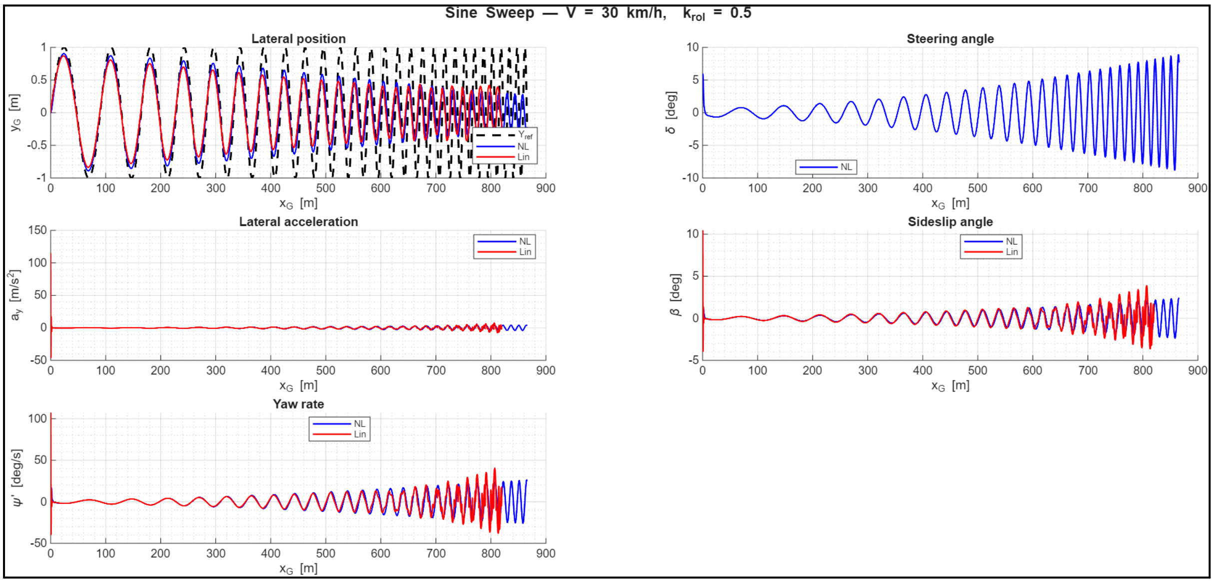
A. Double Lane Change

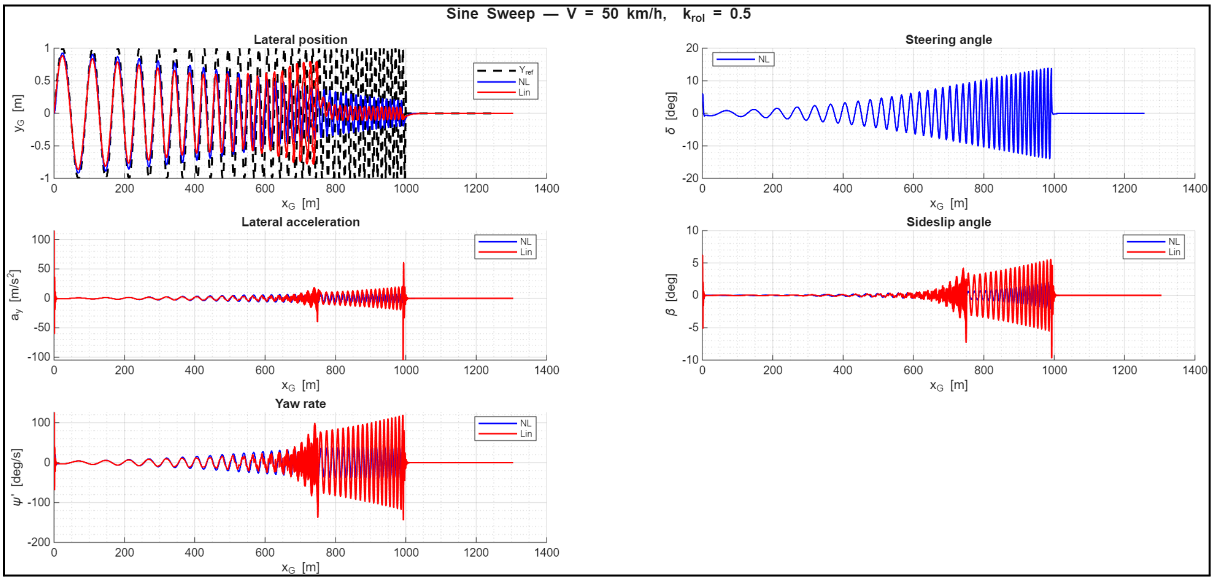


The double lane change results highlight the influence of both vehicle speed and tyre nonlinearity on transient handling performance. At lower speeds, the linear and nonlinear models show similar responses, indicating operation within the linear tyre region. As speed increases, the differences become more pronounced, particularly in yaw rate and sideslip angle, where the nonlinear model exhibits larger peaks and delayed responses due to tyre force saturation. The vehicle remains capable of completing the manoeuvre, but the nonlinear model predicts reduced stability margins and stronger transient effects at higher speeds, demonstrating the importance of nonlinear tyre behaviour during aggressive steering manoeuvres.

B. Sine Sweep







The sine sweep results show that at low excitation frequencies both the linear and nonlinear models exhibit similar responses, indicating that the vehicle operates within the linear tyre region. As the steering frequency increases, the response amplitudes of yaw rate, sideslip angle, and lateral acceleration grow, revealing the approach to the vehicle's lateral dynamic resonance.

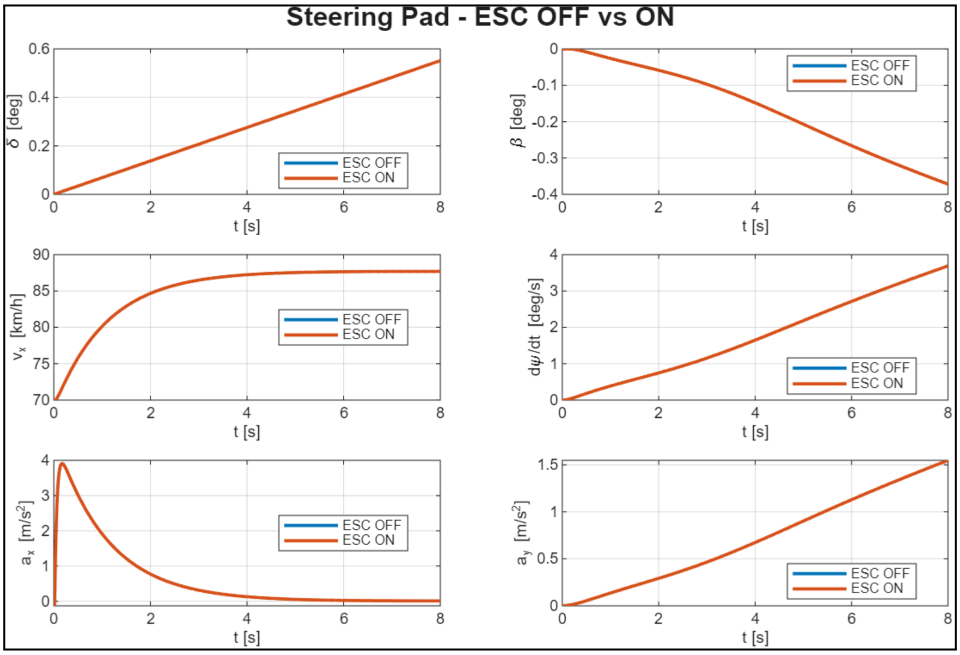
At **30 km/h**, the nonlinear model remains close to the linear prediction throughout most of the manoeuvre, with only small deviations appearing at the highest excitation frequencies. This suggests that tyre forces remain largely within the linear operating range.

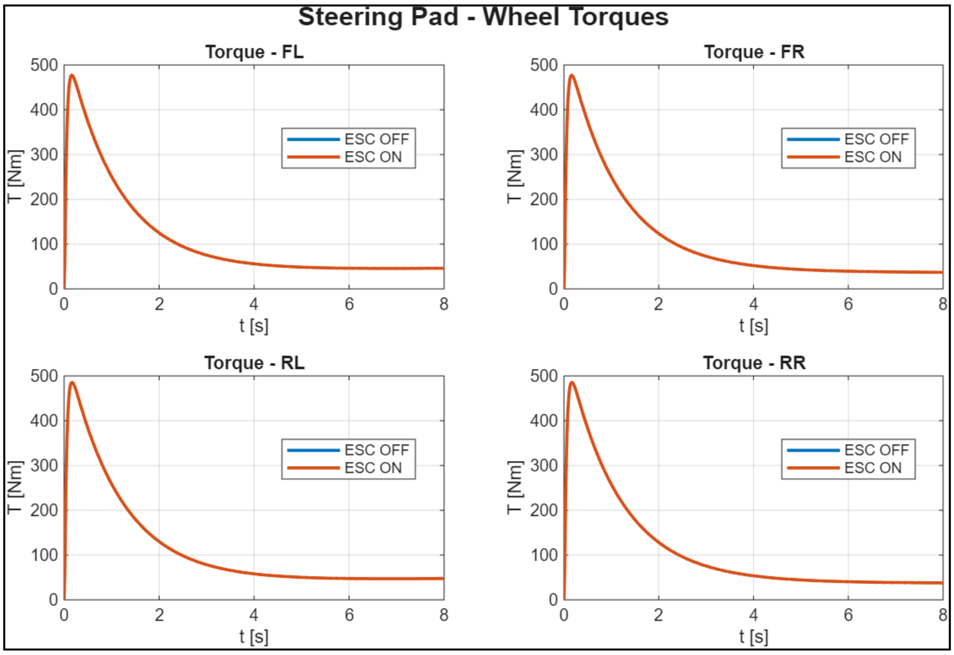
At **50 km/h**, significant differences emerge between the linear and nonlinear responses as the excitation frequency increases. The nonlinear model exhibits reduced response amplitudes and phase shifts compared to the linear model due to tyre force saturation and nonlinear tyre characteristics. These effects become particularly evident near resonance, where the linear model tends to overpredict the vehicle response.

Overall, the results demonstrate that nonlinear tyre behaviour becomes increasingly important as vehicle speed and excitation frequency increase, leading to larger discrepancies between linear and nonlinear vehicle dynamics predictions.

PART 4 - TORQUE VECTORING

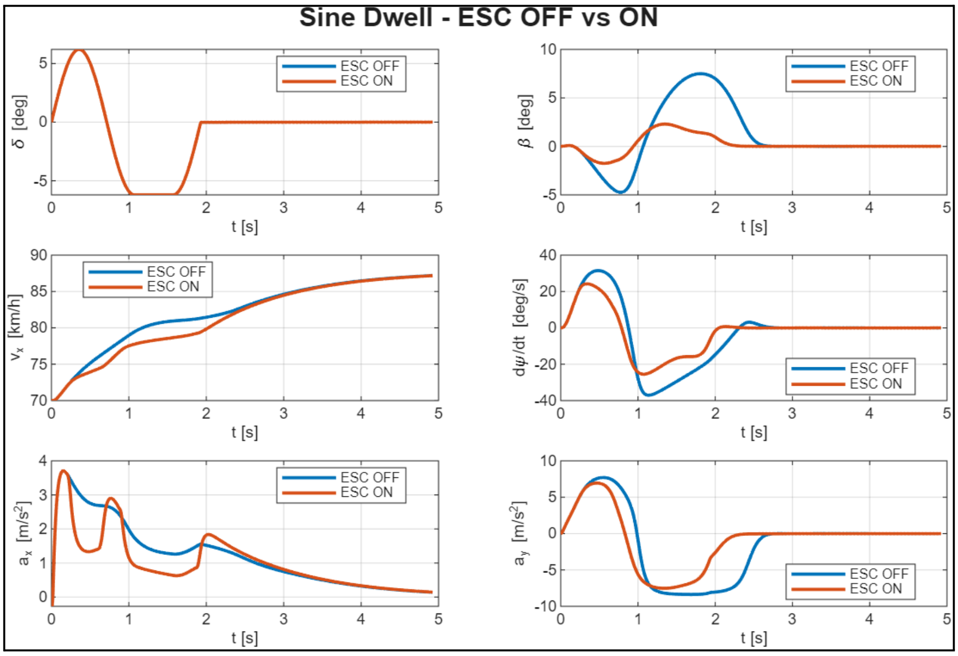
A. Steering Pad (two cases ESC off and ESC on)

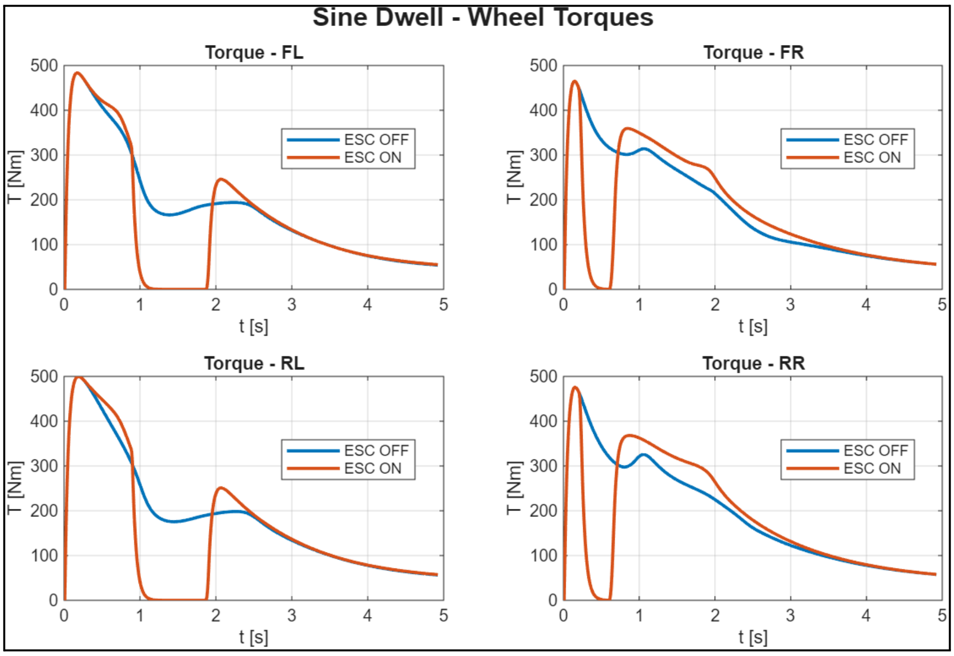




The results show negligible differences between the ESC ON and ESC OFF cases. Vehicle states such as lateral acceleration, yaw rate, sideslip angle, and longitudinal velocity remain nearly identical throughout the manoeuvre. Similarly, the wheel torque histories overlap completely, indicating that no corrective ESC action was required. This behaviour is expected since the steering pad test represents a stable steady-state cornering condition that does not exceed the vehicle's stability limits or trigger ESC intervention.

B. Sine Dwell (two cases ESC off and ESC on)





The results demonstrate the effectiveness of the Electronic Stability Control (ESC) system during the sine dwell manoeuvre. With ESC enabled, both the peak sideslip angle and yaw rate are significantly reduced, resulting in improved vehicle stability and faster recovery following the steering reversal. The wheel torque distributions show active torque modulation during critical phases of the manoeuvre, generating corrective yaw moments that limit excessive vehicle rotation. Consequently, the ESC-equipped vehicle exhibits a more controlled and stable response compared to the ESC OFF case.